

POME Unit - 4 (Part 1 + Part 2)

📘 What Is Economics & Why Is It Important?

🌍 Meaning & Origin

- **Economics** → from Greek *"one who manages a household."*
- It's a **social science** that studies **how humans make decisions** in the face of **scarcity**.
- Decisions can be at **individual, family, business, or societal** levels.

⚖️ Scarcity – The Core of Economics

- **Scarcity** = human wants **>** available resources.
- **Resources** (land, labor, tools, materials) are **limited**, while **human wants** are **unlimited**.
- Even **time** is a scarce resource (only 24 hours/day).
- → Hence, **choice** becomes essential: how to use limited resources efficiently.

🧩 Definitions of Economics

Different economists define it differently:

Perspective	Focus
Wealth Definition	Focus on wealth creation
Welfare Definition	Focus on human welfare
Scarcity Definition	Focus on limited resources
Growth Definition	Focus on economic development

Formal Definition:

> Economics is a social science that studies the efficient allocation of scarce resources to achieve maximum satisfaction of unlimited human needs.

💡 Key Takeaways from the Definition

- Studies **scarce resources**
- Focuses on **allocation**
- Allocation should be **efficient**
- **Human wants** are **unlimited**
- Goal → **maximize satisfaction** with limited means

🧠 Rationales of Economics

Two fundamental facts:

1. **Human wants are unlimited**
2. **Resources are limited (scarce)**

→ Leads to the **basic economic problem**: **Scarcity and Choice**

Economics studies how people **choose** among alternatives to satisfy needs efficiently.

Economists also study outcomes based on:

- **Efficiency**
- **Equity**
- **Stability**
- **Living standards**

🔍 Scope of Economics (WITED)

Economics covers many modern branches like:

- **Development economics**
- **Industrial economics**
- **Transport economics**
- **Welfare economics**
- **Environmental economics**

But the **core branches** are:

🧮 1. Microeconomics

Focus: **Individual units** — households, firms, markets.

- Studies **price determination** & **resource allocation**.
- Tools: **Demand & Supply**.
- Solves: *What, How, and For Whom to produce*.
- Examples: individual income, consumption, savings.

💰 2. Macroeconomics

Focus: **Economy as a whole** — aggregates like national income, output, and employment.

- Studies **income & employment levels**.
- Tools: **Aggregate demand & supply**.
- Solves: *Full employment of resources*.
- Examples: national income, inflation, aggregate consumption.

⚖️ Comparison Table

Microeconomics	Macroeconomics
Studies **individual units**	Studies **whole economy**
Focus: **individual prices, output**	Focus: **national income, price level**
Main issue: **price & resource allocation**	Main issue: **income & employment**
Tools: **demand & supply**	Tools: **aggregate demand & supply**
Concerned with **firm/consumer equilibrium**	Concerned with **national equilibrium**
Examples: individual income, firm output	Examples: national income, inflation

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🧷 Note

> Micro and Macro economics are complementary, not separate — both help in understanding how the economy functions overall.

📘 Economic Analysis & Core Concepts

⚖️ 1. Positive vs. Normative Economics

Aspect	Positive Economics	Normative Economics
Meaning	Describes *"what is"* — based on **facts**	Describes *"what ought to be"* — based on **value judgments**
Nature	**Objective**, fact-based	**Subjective**, opinion-based
Verification	Can be tested or proved using data	Cannot be tested; based on beliefs
Focus	Describes & explains economic behavior	Prescribes policies & evaluates outcomes
Examples	"Inflation rate is 12%.""Unemployment is high."	"The poor should not pay taxes.""Government should control prices."

🧠 **Tip:**

> Positive → Fact-based (is)
>
> Normative → *Opinion-based (ought to be)*

🧩 2. Methods of Economic Reasoning

Economists use **logic** to form theories and analyze behavior — mainly through **two approaches** 👇

🔹 A. Inductive Reasoning

From particular to general (Facts → Theory)

- Based on **observation, data & evidence**.
- Builds theories **after analyzing real facts**.

Steps:

1. Select a problem
2. Collect, classify & analyze data
3. Establish **cause–effect relationships**

🧠 *Example:* Studying real markets → generalize **law of demand**.

🔹 B. Deductive Reasoning

From general to particular (Theory → Facts)

- Begins with **assumptions or principles** and uses **logic** to derive conclusions.

- Theories are **tested later** with real data.

Steps:

1. Identify problem
2. State assumptions
3. Form hypotheses
4. Test validity

🧠 *Example:* Assume "consumers act rationally" → predict market behavior.

💡 3. Scarcity, Choice & Opportunity Cost

🌍 A. Scarcity

- The **central problem** in economics.
- Means **resources are limited**, but **wants are unlimited**.
- Creates a **need for choice** and efficient use.

Scarce Resources Include:

- 👷 **Labour** → human effort → *wage*
- 🌾 **Land** → natural resources → *rent*
- ⚙️ **Capital** → machines, tools → *interest*
- 💼 **Entrepreneurship** → organization & risk-taking → *profit*

🔹 **Free Resource:** abundant at zero price (e.g. sunlight)

🔹 **Economic Resource:** limited relative to demand

📌 **Scarcity ≠ Shortage:**

- *Scarcity* → permanent & universal
- *Shortage* → temporary lack at current price

🔄 B. Choice

- Because of scarcity, we must **choose** what to produce and what to give up.
- Every choice involves **trade-offs** → leads to **opportunity cost**.

💰 C. Opportunity Cost

> The value of the next best alternative forgone to obtain something.

- Measured in **goods/services**, not money.
- Reflects **sacrifice** and **decision-making cost**.

Example:

If 1 meter of cloth = 20 computers,

then producing 1 cloth → opportunity cost = 20 computers.

Principle:

> When opportunity cost rises, people substitute other options instead.

📘 Microeconomics & Macroeconomics

🌍 Overview

Economics studies how people and societies make decisions about **production, distribution, and consumption** of goods and services.

It explores topics like:

- 🏭 Environmental pollution
- 🎓 Role of education in skill development
- 🏢 Behavior of firms, unions & labor
- 💰 Government spending, taxes & regulation

All these topics are studied through **two main branches** 👇

⚖️ 1. Microeconomics vs. Macroeconomics

Aspect	Microeconomics	Macroeconomics
Meaning	Studies **individual units** like households, firms, markets	Studies **the economy as a whole**
Focus	Prices, costs, demand–supply, output, labor	Growth, inflation, unemployment, income
Questions Answered	- What to produce?- How much?- At what price?- How many workers to hire?	- What determines national income?- What causes inflation?- Why do

Aspect	Microeconomics	Macroeconomics
		recessions occur?- How can we ensure full employment?
Tools	Demand, Supply, Consumer & Producer Theory	Fiscal & Monetary Policy
Examples	How firms set prices or households spend	How government reduces inflation
Goal	Efficiency in resource use	Stability & growth in the economy

🧠 **Key Idea:**

> Micro = Individual focus
>
> Macro = *Economy-wide focus*
>
> Both are **interdependent & complementary** — micro decisions shape macro outcomes and vice versa.

🧪 2. Interdependence Example

Think of the economy like a **lake ecosystem** 🌊

- **Micro view:** Study a specific fish or plant (individual firms, consumers).
- **Macro view:** Study the entire lake's balance (national economy).

  Both are vital to understand the whole picture.

💡 **Example:**

- If the **economy grows** → firms **hire more** workers (macro → micro).
- If **firms invest & produce more** → boosts **national output** (micro → macro).

🧮 3. Microeconomics – Key Questions

- How do **households** decide to spend or save?
- How do **firms** decide what and how much to produce?
- How are **prices** and **wages** determined?
- How do **markets** allocate resources efficiently?
- What happens when **markets fail** (e.g., monopoly, pollution)?

🎯 **Focus Areas:**

- Consumer behavior
- Theory of the firm
- Market structure (perfect & imperfect competition)
- Resource markets (labor, capital)

💰 4. Macroeconomics – Key Questions

- What determines **national output (GDP)**?
- Why does **unemployment** occur?
- What causes **inflation** or **deflation**?
- How does the economy **grow** long-term?
- How do **government policies** affect the economy?

🎯 **Main Goals:**

1. **Economic growth**
2. **Low unemployment**
3. **Stable prices (low inflation)**

🏛️ **Policy Tools:**

Policy Type	Controlled By	Focus
Fiscal Policy	Govt. (Congress, Ministry)	Spending & taxation
Monetary Policy	Central Bank	Interest rates, lending, money supply

💭 *Example:* U.S. Federal Reserve → controls monetary policy; Congress → handles fiscal policy.

🧠 5. How Economists Think

> "Economics is a method of thinking, not a fixed doctrine." — John Maynard Keynes

- Economists use **models & theories** to simplify complex realities.
- They assume rational human behavior and analyze relationships between **variables**.

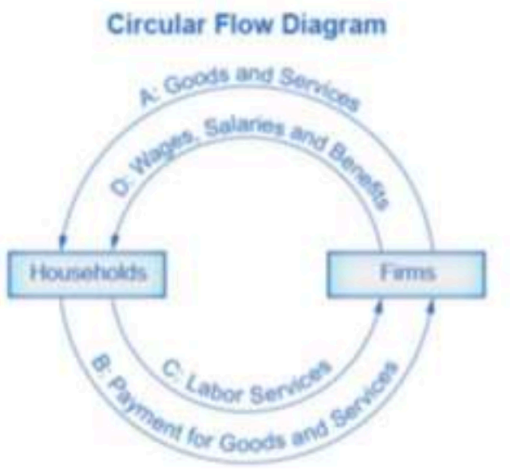
🧩 **Theory:** Abstract explanation of how variables interact.

🧱 **Model:** Practical, simplified representation used to test theories.

🔁 6. The Circular Flow Model

Shows interaction between households and firms in two main markets:

Market Type	Who Supplies?	Who Demands?	What Flows?
Goods & Services Market	Firms	Households	Goods & money
Factor (Labor) Market	Households	Firms	Labor & wages



🌀 Flow Explained:

- **Arrow A:** Firms → Goods/services → Households
- **Arrow B:** Households → Payments → Firms
- **Arrow C:** Households → Labor/resources → Firms
- **Arrow D:** Firms → Wages/rent/profits → Households

📈 **Outer Circle:** Flow of goods & money

⚙️ **Inner Circle:** Flow of labor & wages

> Simplified, but shows the continuous exchange that keeps an economy running.

🧰 7. Theories & Models in Practice

- Economists use models like tools in a **toolbox**.
- They test and refine them with **data**, **graphs**, or **mathematical analysis**.
- Models help:

- Predict outcomes
- Explain real-world behavior
- Evaluate policy impacts

📘 Economic Systems

🌍 Meaning

> Economic System → The way a society organizes and manages production, distribution, and consumption of goods & services to answer basic economic questions:
>
> *What to produce, How to produce, For whom to produce.*

🧩 It's a set of **institutions and arrangements** that shape economic activity.

Three major systems:

1. 🏦 **Capitalist Economy**
2. 🏛️ **Command (Socialist) Economy**
3. ⚖️ **Mixed Economy**

💰 1. Capitalist Economy (Market System / Free Market / Laissez-faire)

🔹 Meaning

- **Private ownership** of all means of production.
- **Profit motive** drives production.
- **Minimal government interference.**

⚙️ Key Features

Feature	Description
Private Property	Individuals own land, factories, machinery, etc.
Consumer Sovereignty	Consumers decide what to produce through their choices.
Profit Motive	Entrepreneurs produce to maximize profits.

Feature	Description
Competition	Exists among sellers, buyers, and workers.
Price Mechanism	Market prices regulate supply & demand.
Limited Govt. Role	Govt. only ensures defense & law and order.
Self-Interest	Each person acts for personal economic gain.
Inequality	Wide gap between rich & poor.
Negative Externalities	Pollution, exploitation — social costs ignored.

✅ Advantages

- Flexible & adaptable
- Decentralized economic power
- High per capita income & living standards
- Encourages innovation & entrepreneurship
- Efficient use of resources
- Rapid capital formation

❌ Disadvantages

- Inequality of income & wealth
- Exploitation of labor
- Regional imbalance
- Negative externalities (e.g. pollution)
- Unbalanced development

🏛️ 2. Command Economy (Socialist Economy)

🔹 Meaning

- **Government owns & controls** all means of production.
- Economic decisions made through **central planning**.
- Aims for **social welfare** and **equal income distribution**.

⚙️ Key Features

Feature	Description
Collective Ownership	No private property — owned by the state.
Central Planning	Govt. allocates resources & sets targets.
Strong Govt. Role	Full control over production, prices, and employment.
Social Welfare Goal	Eliminates exploitation & promotes equality.
Equal Incomes	Little gap between rich and poor.

✅ Advantages

- No wasteful competition
- Balanced regional & sectoral growth
- Eliminates monopolies & inequalities
- Ensures job security & basic needs

❌ Disadvantages

- No automatic price mechanism
- Lack of incentives → inefficiency
- Bureaucracy & red-tapism
- Absence of freedom for producers & consumers
- Slower economic growth

⚖️ 3. Mixed Economy

🔹 Meaning

- Combines the **benefits of capitalism and socialism.**
- Both **private and public sectors** coexist and cooperate.

⚙️ Key Features

Feature	Description
Co-existence of Sectors	Public sector → heavy industries, defense; Private sector → consumer goods.

Feature	Description
Economic Welfare Goal	Govt. regulates private activities for social good.
Economic Planning	Govt. ensures coordinated development.
Controlled Price Mechanism	Prices in public sector regulated; others market-based.
Economic Equality	Progressive taxes, subsidies & property limits prevent inequality.

✅ Advantages

- Combines benefits of both systems
- Ensures freedom + social welfare
- Promotes balanced and rapid development
- Encourages private investment with regulation
- Reduces inequality

❌ Disadvantages

- Inefficiency in public sector
- Govt. control may restrict private initiative
- Possibility of corruption, black markets
- Risk of economic instability if regulation is weak

CIE - 2 STARTS HERE (Part 2)

📘 Demand, Supply & Equilibrium in Markets for Goods and Services

🌍 1. Markets for Goods and Services

A **market** is any arrangement that brings **buyers and sellers** together to exchange goods or services.

Markets coordinate economic activity through **prices**, without the need for central direction.

Economists use the **demand–supply model** to understand:

- How prices are determined
- How quantities exchanged are decided
- Why markets sometimes face **surpluses** or **shortages**

📉 2. Demand

🔹 Meaning of Demand

Demand shows the quantities of a good or service that **buyers are willing and able to purchase** at various prices, **during a given time period**, assuming other factors remain constant.

👉 *Willing* = desire to buy

👉 *Able* = having purchasing power (income)

🔹 Law of Demand

> Other things remaining constant, as price falls, quantity demanded rises; as price rises, quantity demanded falls.

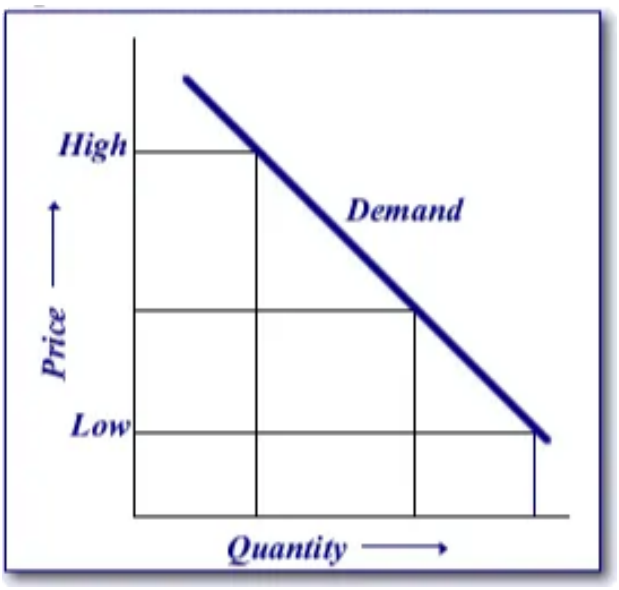
📌 This inverse relationship exists because of:

- **Substitution effect** – consumers switch to cheaper alternatives
- **Income effect** – lower prices increase real purchasing power

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🔹 Demand Curve

- **Downward sloping** from left to right
- Shows the **price–quantity relationship**
- A **change in price** causes **movement along the demand curve**, not a shift



📈 3. Supply

🔹 Meaning of Supply

Supply shows the quantities of a good or service that **sellers are willing and able to offer for sale** at various prices, during a given time period.

🔹 Law of Supply

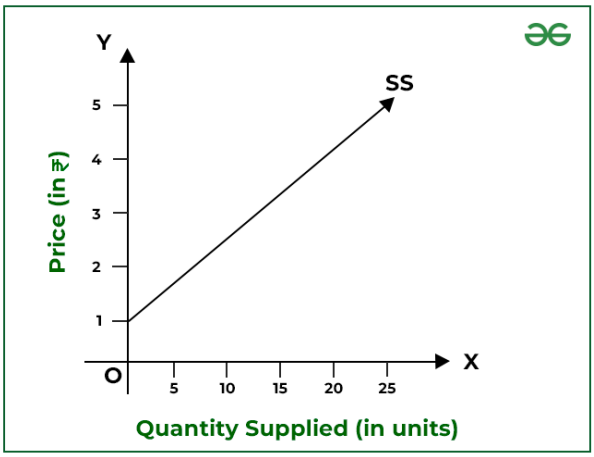
> Other things remaining constant, as price rises, quantity supplied rises; as price falls, quantity supplied falls.

📌 This direct relationship exists because:

- Higher prices increase **profit incentives**
- Producers are willing to expand output at higher prices

🔹 Supply Curve

- **Upward sloping** from left to right
- Shows how quantity supplied responds to price changes
- A **change in price** leads to **movement along the supply curve**, not a shift



⚖️ 4. Market Equilibrium

🔹 Meaning of Equilibrium

Market equilibrium occurs when:

> Quantity demanded equals quantity supplied.

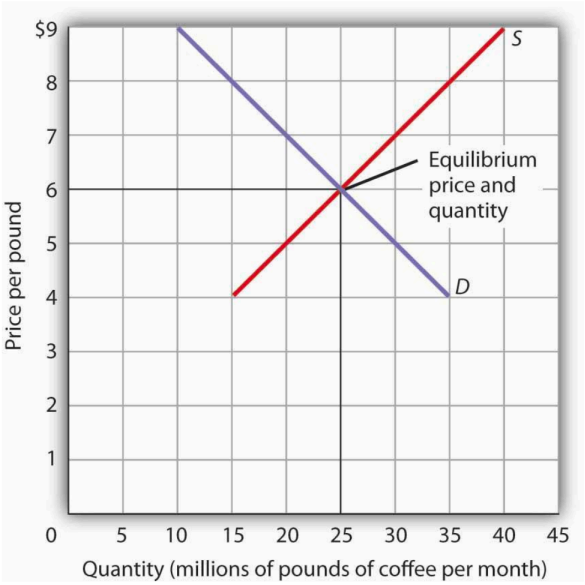
At this point:

- Buyers can buy exactly what they want
- Sellers can sell exactly what they want
- There is **no pressure for price to change**

🔹 Determination of Equilibrium Price and Quantity

By **combining demand and supply curves** on the same graph:

- The **intersection point** gives:
  - **Equilibrium Price**
  - **Equilibrium Quantity**



📊 Example: Coffee Market

- **Equilibrium Price = $6 per pound**
- **Equilibrium Quantity = 25 million pounds per month**

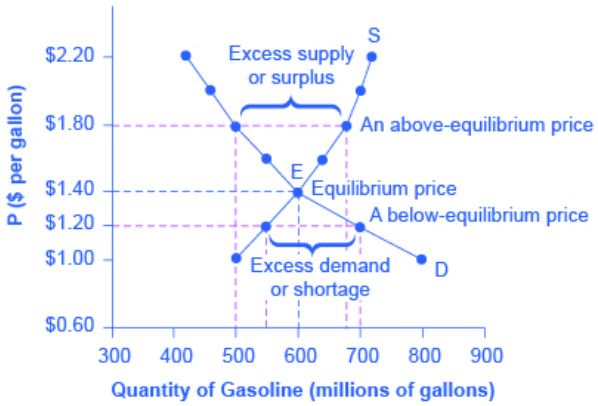
At this price:

- Quantity demanded = Quantity supplied
- The market is in **balance**

🧠 **Key idea:**

> Unless demand or supply shifts, the equilibrium price will remain stable.

📦 5. Surplus (Excess Supply)



🔹 Meaning of Surplus

A **surplus** occurs when:

> Quantity supplied exceeds quantity demanded at a given price.

📌 This happens when **price is above the equilibrium price**.

🔹 Example: Surplus in Coffee Market

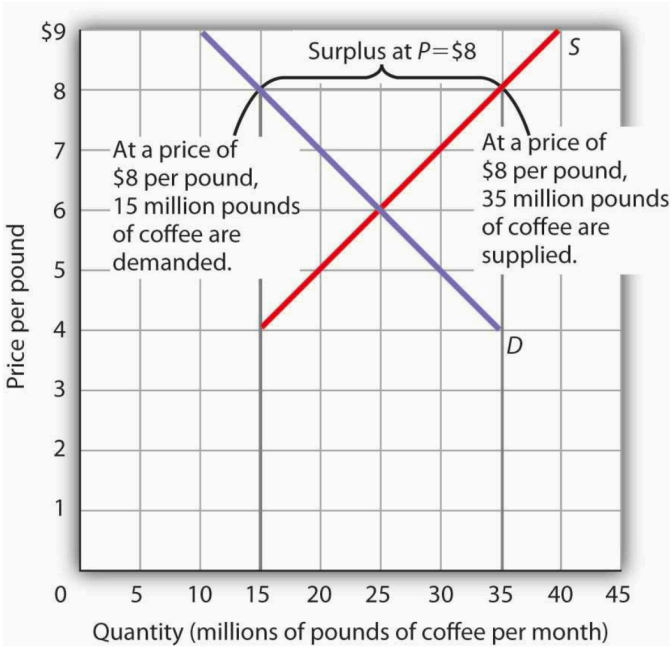
- Price = **$8 per pound**
- Quantity demanded = **15 million pounds**
- Quantity supplied = **35 million pounds**
- **Surplus = 20 million pounds**

🔹 Market Adjustment under Surplus

- Unsold goods accumulate
- Sellers **reduce prices** to clear excess stock
- As price falls:
  - Quantity demanded **increases**
  - Quantity supplied **decreases**
- Process continues until **equilibrium is restored**

🧠 **Important Note:**

- Changes occur due to **price changes**, not shifts in curves
- Surpluses are generally **short-lived**



🚨 6. Shortage (Excess Demand)

🔹 Meaning of Shortage

A **shortage** occurs when:

> Quantity demanded exceeds quantity supplied at a given price.

📌 This happens when **price is below the equilibrium price**.

🔹 Example: Shortage in Coffee Market

- Price = **$4 per pound**
- Quantity demanded = **35 million pounds**
- Quantity supplied = **15 million pounds**
- **Shortage = 20 million pounds**

🔹 Market Adjustment under Shortage

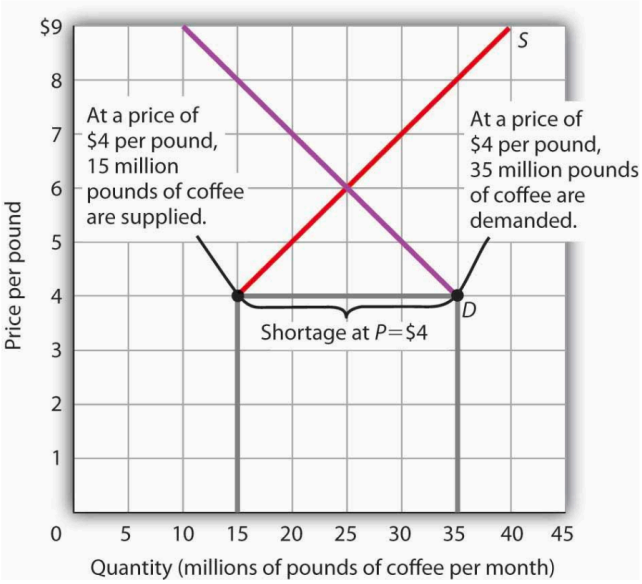
- Consumers compete to buy limited goods
- Sellers respond by **raising prices**
- As price rises:

- Quantity supplied **increases**
- Quantity demanded **decreases**

- Adjustment continues until **equilibrium price is reached**

🧠 **Key insight:**

> Shortages push prices upward toward equilibrium.



🔄 7. Price Changes vs Curve Shifts

Situation	What Changes?	Result
Change in price	Price only	Movement **along** demand or supply curve
Change in demand or supply conditions or anything else	Curve position	**Shift** of curve

📌 In surplus and shortage analysis:

- **Demand and supply curves do not shift**
- Only **price adjusts**

📘 Shifts in Demand and Supply

🌍 1. What Do We Mean by "Shifts"?

In demand–supply analysis, a **shift** means:

> A change in the entire demand curve or supply curve, caused by factors other than price.

📌 This is different from:

- **Movement along a curve** → caused by **price change**
- **Shift of a curve** → caused by **non-price determinants**

A shift always leads to a **new equilibrium price and quantity**.

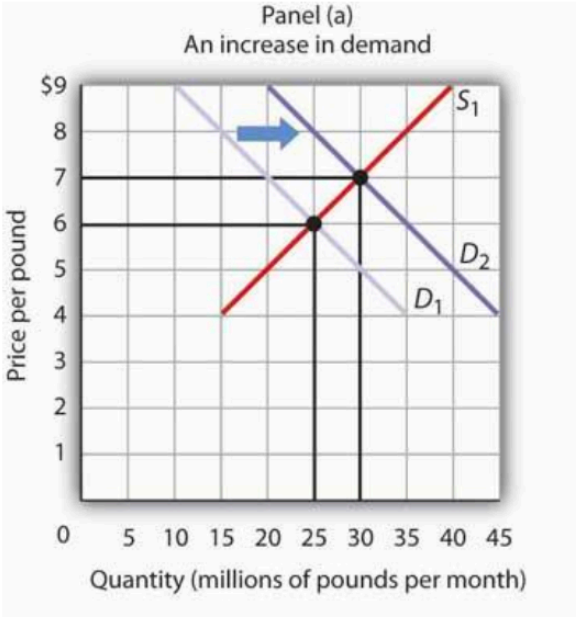
📉 2. Shifts in Demand

🔹 Increase in Demand (Rightward Shift)

An **increase in demand** means that **at every price**, buyers want to purchase **more** than before.

📊 Graphically

- Demand curve shifts **rightward** ($D_1 \rightarrow D_2$)
- Supply curve remains unchanged
- New intersection → **higher equilibrium price & higher equilibrium quantity**



📌 Example (Coffee Market)

- Original equilibrium:
  - Price = **$6 per pound**
  - Quantity = **25 million pounds**
- After increase in demand:
  - New price = **$7 per pound**
  - New quantity = **30 million pounds**

🧠 **Key insight:**

> Price rises → causes a movement along the supply curve, not a supply shift.

🔹 Causes of Increase in Demand

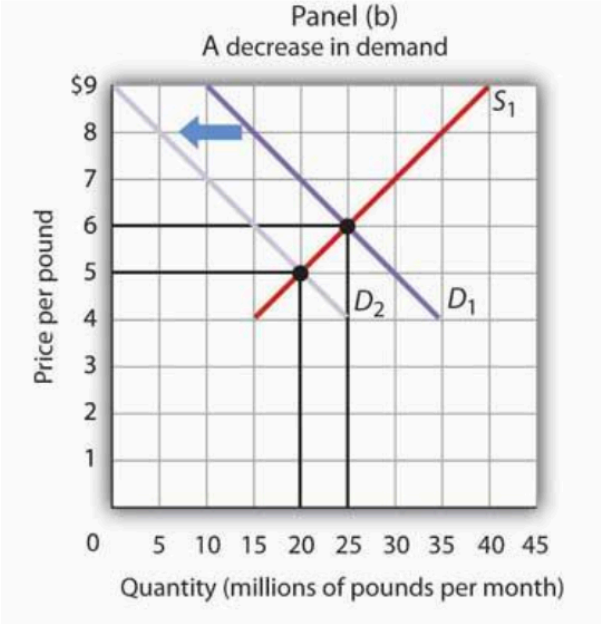
- Change in **tastes/preferences** (more coffee lovers ☕)
- Increase in **income** (for normal goods)
- Increase in **population**
- Increase in price of a **substitute** (tea becomes expensive)
- Decrease in price of a **complement** (doughnuts cheaper)
- **Expectations** of higher future prices

🔹 Decrease in Demand (Leftward Shift)

A **decrease in demand** means that **at every price**, buyers want **less** than before.

📊 Graphically

- Demand curve shifts **leftward**
- Supply curve unchanged
- New equilibrium → **lower price & lower quantity**



📌 Example (Coffee Market)

- New equilibrium price = **$5 per pound**
- Quantity supplied falls to **20 million pounds per month**

🔹 Causes of Decrease in Demand

- Change in tastes away from coffee
- Decrease in income
- Decrease in population
- Decrease in price of substitutes
- Increase in price of complements
- Expectation of **lower future prices**

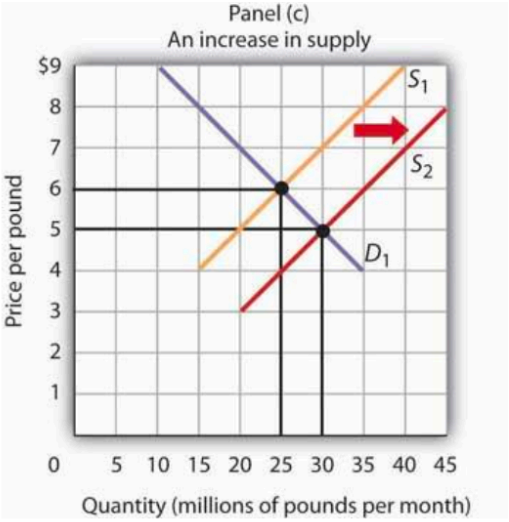
📈 3. Shifts in Supply

🔹 Increase in Supply (Rightward Shift)

An **increase in supply** means that **at every price**, sellers are willing to sell **more**.

📊 Graphically

- Supply curve shifts **rightward** ($S_1 \rightarrow S_2$)
- Demand curve unchanged
- New equilibrium → **lower price & higher quantity**



📌 Example (Coffee Market)

- New equilibrium price = **$5 per pound**
- Quantity demanded increases to **30 million pounds per month**

🧠 **Key insight:**

> Price falls → causes a movement along the demand curve, not a demand shift.

🔹 Causes of Increase in Supply

- Decrease in **input prices** (labor, fertilizers)
- Improvement in **technology**
- Favorable **weather conditions**

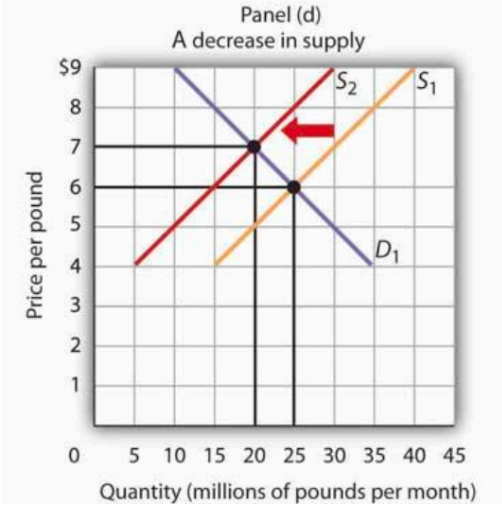
- Increase in number of **firms**
- Lower returns from alternative uses of inputs

🔹 Decrease in Supply (Leftward Shift)

A **decrease in supply** means that **at every price**, sellers are willing to sell **less**.

📊 Graphically

- Supply curve shifts **leftward**
- Demand curve unchanged
- New equilibrium → **higher price & lower quantity**



📌 Example (Panel d – Coffee Market)

- Supply curve shifts left
- New equilibrium price = **$7 per pound**
- Quantity demanded falls to **20 million pounds per month**

🔹 Causes of Decrease in Supply

- Increase in **input prices**
- Higher returns from **alternative uses** of inputs
- Technological problems (e.g. pesticide restrictions)

- Reduction in number of firms
- Natural events (floods, excessive rain, drought)

🔄 4. Important Distinction: Shift vs Movement

Aspect	Movement Along Curve	Shift of Curve
Cause	Change in price	Change in non-price factors
Curve	Same curve	New curve
Example	Price rises → Qs increases	Technology improves → supply increases

📌 **Surpluses & shortages** are resolved by **movements**,

📌 **New equilibria** are created by **shifts**.

🧠 5. Combined Understanding

- **Demand shifts** change equilibrium by affecting buyers' behavior
- **Supply shifts** change equilibrium by affecting sellers' costs & capacity
- Market equilibrium continuously adjusts due to changes in:
  - Preferences
  - Technology
  - Income
  - Input prices
  - Expectations
  - Natural conditions

📘 Elasticity of Demand and Supply

🌍 1. Meaning of Elasticity

Elasticity measures the **degree of responsiveness** of one economic variable to a change in another.

In markets, elasticity answers questions like:

- How much does **quantity demanded** change when **price or income** changes?
- How much does **quantity supplied** change when **price** changes?

📌 Elasticity focuses on **percentage changes**, not absolute changes — this allows meaningful comparison.

📉 2. Elasticity of Demand

🔹 Definition

Elasticity of demand measures how responsive the **quantity demanded of a good** is to changes in:

- Its **price**
- Consumer **income**
- Prices of **related goods**

🧮 3. Price Elasticity of Demand (PED)

🔹 Meaning

Price Elasticity of Demand (PED) measures the responsiveness of quantity demanded to a change in **price**, other things remaining constant.

🔹 Formula

$$\text{PED} = \frac{\%\text{ change in quantity demanded}}{\%\text{ change in price}}$$

📌 PED is usually **negative** because price and quantity demanded move in opposite directions

(but we often ignore the negative sign for simplicity).

🔹 Types of Price Elasticity of Demand

Type	Value of PED	Meaning
Elastic Demand	PED > 1	Quantity demanded changes **more than proportionately** to price
Inelastic Demand	PED < 1	Quantity demanded changes **less than proportionately**
Unitary Elastic	PED = 1	Proportionate change in price and quantity
Perfectly Elastic	PED = ∞	Infinite demand at one price
Perfectly Inelastic	PED = 0	Quantity demanded does not change at all

🔹 Interpretation

- **Elastic demand** → consumers are **price sensitive**
- **Inelastic demand** → consumers are **price insensitive**

🧠 **Examples:**

- Salt, medicines → inelastic
- Luxury goods, branded electronics → elastic

💰 4. Income Elasticity of Demand (YED)

🔹 Meaning

Income Elasticity of Demand (YED) measures how quantity demanded responds to a change in **consumer income**.

🔹 Formula

$$\text{YED} = \frac{\%\text{ change in quantity demanded}}{\%\text{ change in income}}$$

🔹 Types Based on YED

Type of Good	YED Value	Explanation
Normal Goods	YED > 0	Demand rises as income rises
Luxury Goods	YED > 1	Demand rises more than income
Necessities	0 < YED < 1	Demand rises slowly with income

Type of Good	YED Value	Explanation
Inferior Goods	YED < 0	Demand falls as income rises

🧠 **Examples:**

- Luxury cars → high positive YED
- Basic food items → low positive YED
- Cheap substitutes → negative YED

📈 5. Elasticity of Supply

🔹 Definition

Elasticity of supply measures how responsive the **quantity supplied** of a good is to changes in **price**, other factors remaining constant.

🧮 6. Price Elasticity of Supply (PES)

🔹 Formula

$$\text{PES} = \frac{\%\text{ change in quantity supplied}}{\%\text{ change in price}}$$

🔹 Types of Price Elasticity of Supply

Type	Value of PES	Meaning
Elastic Supply	PES > 1	Quantity supplied responds significantly
Inelastic Supply	PES < 1	Quantity supplied responds weakly
Unitary Elastic	PES = 1	Proportionate response
Perfectly Elastic	PES = ∞	Infinite supply at a given price
Perfectly Inelastic	PES = 0	Quantity supplied is fixed

🔍 7. Determinants of Elasticity

🔹 Determinants of Demand Elasticity (TINS)

1. **Availability of Substitutes**
    - More substitutes → **more elastic** demand
    - Fewer substitutes → **inelastic**
2. **Nature of the Good**
    - Necessities → inelastic
    - Luxuries → elastic
3. **Proportion of Income Spent**
    - Higher cost relative to income → more elastic
4. **Time Period**
    - Short run → inelastic
    - Long run → more elastic (adjustment possible)

🔹 Determinants of Supply Elasticity

1. **Production Flexibility**
    - Easily scalable production → elastic supply
2. **Availability of Resources**
    - Limited or fixed resources → inelastic supply
3. **Time Frame**
    - Short run → inelastic
    - Long run → elastic

🧠 *Agricultural products* → usually inelastic in short run

🧠 *Manufactured goods* → more elastic in long run

⚖️ 8. Demand vs Supply Elasticity (Quick Comparison)

Aspect	Demand Elasticity	Supply Elasticity
Focus	Buyer response	Seller response
Main factor	Preferences, income	Cost, technology
Time effect	Strong	Very strong
Example	Luxury goods	Industrial goods

📘 Price Elasticity of Demand (PED) & Price Elasticity of Supply (PES)

🌍 1. Meaning of Price Elasticity

Price elasticity measures how **responsive quantity** is to a **change in price**.

- For **demand** → how buyers react to price changes
- For **supply** → how sellers react to price changes

📌 Elasticity is a **ratio of percentage changes**, so it has **no units**.

📉 2. Price Elasticity of Demand (PED)

🔹 Definition

> Price Elasticity of Demand measures the responsiveness of quantity demanded to a change in price, other things remaining constant.

🔹 Formula

$$\text{PED} = \frac{\%\ \text{change in quantity demanded}}{\%\ \text{change in price}}$$

- PED is **negative** (price ↑ → quantity demanded ↓)
- By convention, we use the **absolute value**

⚖️ 3. Types of Price Elasticity (Demand & Supply)

Condition	Elasticity Value	Interpretation
Elastic	> 1	Quantity responds **more than proportionately**
Unitary Elastic	= 1	Proportionate response
Inelastic	< 1	Quantity responds **less than proportionately**

📌 Interpretation logic:

- %ΔQ > %ΔP → **Elastic**
- %ΔQ = %ΔP → **Unitary**
- %ΔQ < %ΔP → **Inelastic**

🧮 4. The Midpoint Method (Very Important)

🔹 Why Midpoint Method?

If we calculate elasticity using only the initial value:

- Price increase → one value
- Price decrease → a different value

👉 To avoid this problem, economists use the **Midpoint Method**, which gives the **same elasticity** regardless of direction.

🔹 **Midpoint Formula**

$$\% \text{ change in quantity} = \frac{Q_2 - Q_1}{(Q_1 + Q_2)/2} \times 100$$

$$\% \text{ change in price} = \frac{P_2 - P_1}{(P_1 + P_2)/2} \times 100$$

Then:

$$\textbf{Elasticity} = \frac{\%\Delta Q}{\%\Delta P}$$

📊 5. Calculating Price Elasticity of Demand (Numerical Example)

🔹 Example: Coffee Market (Figure 4.5)

- Price falls from **$70 to $60**
- Quantity increases from **2,800 to 3,000 units**

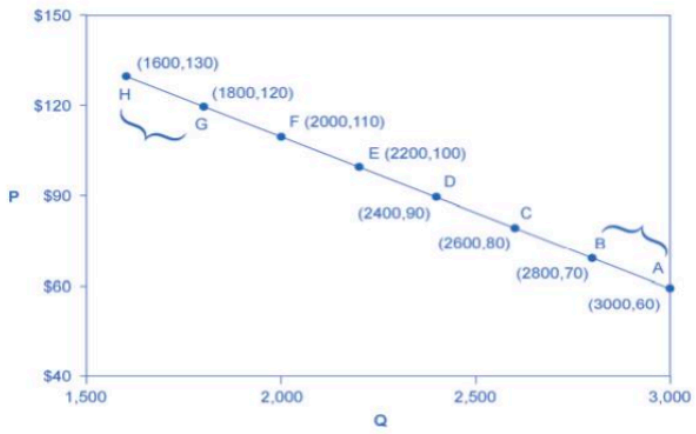


Fig 4.5 We calculate the price elasticity of demand as the ratio of percentage change in quantity to percentage change in price

First, apply the formula to calculate the elasticity as price decreases from \$70 at point B to \$60 at point A:

$$\begin{aligned}
\%\text{ change in quantity} &= \frac{3{,}000 - 2{,}800}{(3{,}000 + 2{,}800)/2} \times 100 \\
&= \frac{200}{2{,}900} \times 100 \\
&= 6.9 \\
\%\text{ change in price} &= \frac{60 - 70}{(60 + 70)/2} \times 100 \\
&= \frac{-10}{65} \times 100 \\
&= -15.4 \\
\text{Price Elasticity of Demand} &= \frac{6.9\%}{-15.4\%} \\
&= 0.45
\end{aligned}$$

🔹 Interpretation

- PED = **0.45 (< 1)**
- Demand is **inelastic**
- A **1% change in price** causes only a **0.45% change in quantity demanded**

📌 Example interpretation:

- 10% increase in price → only 4.5% fall in quantity demanded

---

📈 6. Price Elasticity of Supply (PES)

🔹 Definition

> Price Elasticity of Supply measures the responsiveness of quantity supplied to a change in price.

🔹 Formula

$$\text{PES} = \frac{\%\text{ change in quantity supplied}}{\%\text{ change in price}}$$

- PES is **positive**
- Read as an **absolute value**

📊 7. Calculating Price Elasticity of Supply (Numerical Example)

🔹 Example: Apartment Rentals (Figure 4.6)

- Rent rises from **$650 to $700**
- Quantity supplied rises from **10,000 to 13,000 units**

Step 1: % Change in Quantity

$$\frac{13,000 - 10,000}{(13,000 + 10,000)/2} \times 100 = \frac{3,000}{11,500} \times 100 = 26.1\%$$

Step 2: % Change in Price

$$\frac{700 - 650}{(700 + 650)/2} \times 100 = \frac{50}{675} \times 100 = 7.4\%$$

Step 3: PES

$$PES = \frac{26.1}{7.4} = 3.53$$

🔹 Interpretation

- PES = **3.53 (> 1)**
- Supply is **elastic**

- A **1% rise in price** causes a **3.5% rise in quantity supplied**

🧠 8. Elasticity vs Slope (Common Confusion)

Aspect	Elasticity	Slope
Measures	Responsiveness (%)	Rate of change
Depends on	Units + position	Units only
Varies along curve	Yes	No (straight line)

📌 Elasticity changes **along the curve**, even if slope is constant.

📊 CHANGES IN INCOME AND PRICES AFFECTING CONSUMPTION CHOICES

🧠 Key Concepts

🔹 Utility

- **Utility** = satisfaction gained from consumption
- Consumers aim to **maximize total utility** given limited income

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🔹 Budget Constraint

- Represents **all combinations of goods** a consumer can afford
- Determined by:
  - Income
  - Prices of goods
- Acts as a **limit** on consumption choices

💰 Impact of Income Changes

🔹 Normal Goods

- Demand **increases when income rises**
- Examples:

- Organic food
- Better housing
- Luxury items

🔹 Inferior Goods

- Demand **decreases when income rises**
- Consumers shift to better alternatives
- Examples:
  - Instant noodles
  - Used clothes

🔹 Income Effect

- Change in income alters **purchasing power**
- Increase in income → demand curve shifts **rightward**
- Decrease in income → demand curve shifts **leftward**

💸 Impact of Price Changes

🔹 Substitution Effect

- When price of a good **falls**, it becomes cheaper relative to substitutes
- Consumers substitute:
  - **More of cheaper good**
  - **Less of costlier alternatives**

📌 Example:

If chicken becomes cheaper than beef → consumers buy more chicken

---

🔹 Income Effect (Price-related)

- Fall in price makes consumers feel **richer**
- Same income can now buy **more goods**
- Leads to higher consumption

🔹 Demand Curve Response

- Price ↓ → Quantity demanded ↑
- Price ↑ → Quantity demanded ↓

📌 Note:

- Price change causes **movement along the curve**, not a shift

---

📈 INDIFFERENCE CURVES & CONSUMER CHOICE

🔹 Indifference Curves

- Show combinations of two goods giving **equal satisfaction**
- Higher curve → higher utility
- Never intersect

🔹 Optimal Consumption Point

- Occurs where:
  - **Budget line is tangent** to an indifference curve
- Represents **maximum utility** achievable with given income & prices

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🌍 REAL-WORLD APPLICATIONS

🏛️ Policy Implications

- Income support / stimulus → boosts consumption
- Taxes raising prices → may reduce demand temporarily
- Helps in **welfare and taxation policy design**

🧪 Market Research

- Firms analyze income & price sensitivity

- Helps decide:
  - Pricing strategies
  - Advertising budgets
  - Product positioning

📘 What do MR and MC stand for?

🔹 MR = Marginal Revenue

> The **additional revenue** a firm earns from selling **one more unit** of output.

Formula:

$$MR = \frac{\Delta TR}{\Delta Q}$$

Where:

- **TR** = Total Revenue
- **Q** = Quantity

📌 In simple words:

If selling one extra unit increases revenue by ₹50, then **MR = ₹50**.

🔹 MC = Marginal Cost

> The **additional cost** of producing **one more unit** of output.

Formula:

$$MC = \frac{\Delta TC}{\Delta Q}$$

Where:

- **TC** = Total Cost
- **Q** = Quantity

📌 Example:

If producing one more unit increases cost by ₹30, then **MC = ₹30**.

🎯 Why Are MR and MC Important?

Firms **maximize profit** where:

$$MR = MC$$

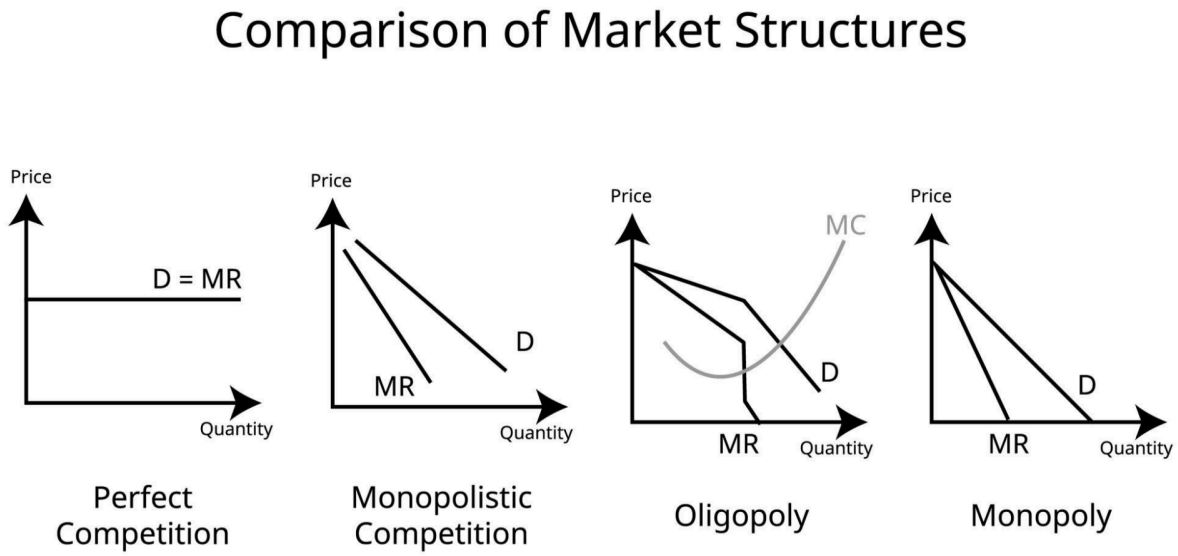
🧠 Why?

- If **MR > MC** → Producing more increases profit
- If **MR < MC** → Producing more reduces profit
- If **MR = MC** → Profit is maximized

This rule applies to:

- Perfect competition
- Monopoly
- Monopolistic competition
- Oligopoly

📘 Four Market Structures in Economics



🌍 1️⃣ Perfect Competition

🔹 Meaning

A market structure with **many buyers and sellers**, selling **identical products**, where **no firm can influence price**.

Firms are **price takers**.

🔹 Key Features

- Large number of firms
- Homogeneous products
- Free entry and exit
- Perfect information
- No selling costs

🔹 Short-Run Equilibrium

- Profit maximized where **MR = MC**
- Since firm is price taker:

  👉 **P = MR = AR**

- Firm can earn:
  - Supernormal profit
  - Normal profit
  - Loss

🔹 Long-Run Equilibrium

- **P = MC = ATC (minimum point)**
- Zero economic profit
- No excess capacity
- **Allocatively & Productively efficient**

🧠 Efficiency Insight

✔️ Price = Marginal Cost → Socially optimal

✔️ Output at minimum ATC → No waste

🌍 2️⃣ Monopoly

🔹 Meaning

A market with **one seller**, producing a product with **no close substitutes**.

Firm is a **price maker**.

🔹 Key Features

- Single firm
- High barriers to entry
- Unique product
- Downward-sloping demand curve
- Price > MC

🔹 Equilibrium

- Output where **MR = MC**
- Price determined from demand curve
- **P > MC**
- Can earn supernormal profit in long run

🔹 Inefficiencies

- Allocative inefficiency (P > MC)
- Productive inefficiency
- Deadweight loss
- Lower output, higher price

🧠 Key Insight

Monopoly restricts output to raise price.

🌍 3️⃣ Monopolistic Competition (don't go in depth)

🔹 Meaning

Many firms selling **similar but differentiated products**.

Each firm has **limited price control**.

🔹 Key Features

- Many sellers
- Product differentiation
- Free entry & exit
- Selling & advertising costs
- Downward-sloping demand

🔹 Short Run

- MR = MC
- Can earn profit or incur loss

🔹 Long Run

- Entry of firms shifts demand left
- **AR = ATC**
- Zero economic profit
- Excess capacity exists

🔹 Inefficiencies

- P > MC
- Not at minimum ATC
- Advertising costs
- Allocative & productive inefficiency

🧠 Trade-off

Less efficiency than perfect competition

But more product variety

🌍 4️⃣ Oligopoly

🔹 Meaning

A market dominated by **few large firms**.

Firms are **interdependent** — decisions of one affect others.

🔹 Key Features

- Few dominant firms
- High entry barriers
- Strategic behavior

- Possible collusion
- Non-price competition

🔹 Demand Curve (Kinked Model)

- Price rigidity
- Small price changes may not affect price level
- MR has discontinuity

🔹 Types of Oligopoly

- Collusive (Cartel)
- Non-collusive

🔹 Outcomes

- Price above MC
- Potential inefficiency
- May behave like monopoly (cartel case)

📊 COMPARISON TABLE (Exam Gold)

Feature	Perfect Competition	Monopoly	Monopolistic Competition	Oligopoly
Number of Firms	Many	One	Many	Few
Product	Identical	Unique	Differentiated	Homogeneous/Differentiated
Price Control	None	Full	Limited	Considerable
Demand Curve	Perfectly elastic	Downward sloping	Downward sloping	Kinked/Strategic
Entry Barriers	None	High	Low	High
Long-run Profit	Zero	Possible	Zero	Possible
Efficiency	Fully efficient	Inefficient	Inefficient	Generally inefficient